

PR1.

| Study-hours | attendance | group-discussion | previous-test-score | final-exam-pass |
|-------------|------------|------------------|---------------------|-----------------|
| 12          | 90         | yes              | 85                  | pass            |
| 8           | 75         | no               | 60                  | fail            |
| 15          | 95         | yes              | 91                  | pass            |
| 5           | 65         | no               | 48                  | fail            |
| 10          | 82         | yes              | 72                  | pass            |
| 14          | 88         | yes              | 80                  | fail/pass       |
| 7           | 70         | no               | 55                  | fail            |
| 11          | 84         | yes              | 76                  | pass            |
| 9           | 78         | no               | 67                  | fail            |
| 13          | 92         | yes              | 89                  | pass            |

1. Understanding the basics

→ Explain in your own words:

→ What is probability

- probability is the chance of an event occurring. its value ~~it~~ lies between 0 and 1.

Formula:  $P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$

→ Key probability Terminology

- Experiment: An activity with uncertain outcomes
- Outcome: A possible result of an experiment
- Sample Space: Set of all possible outcomes
- Event: A specific outcome or group of outcomes
- Random Variable: A variable whose value depends on a random experiment

→ These prop. probability Events from the Dataset.

- Event A: Student studies more than 10 hours/week.
- Event B: Student attendance is more than 80%.
- Event C: Student passes the final Exam.

## Q2.1 Types of Event

### 1. Empirical probability.

Based on observed data.

Ex:-

if 138 out of 200 Students pass.

$$P(\text{pass}) = \frac{138}{200} = 0.69$$

Ans:- 0.69 (69%)

### 2. Theoretical probability.

Based on mathematical assumption.

If pass and fail are equally likely,

$$P(\text{pass}) = \frac{1}{2} = 0.50$$

Ans:- 0.50 (50%)

## Q.2) Random Variable ~~are~~ and probability Distribution

### 1. Random Variable

Let  $x$  be the number of students who pass the final exam out of 3 randomly selected students

### 2. Probability Distribution Table

| $x$ (Students passed) | Probability $P(x)$ |
|-----------------------|--------------------|
| 0                     | 0.03               |
| 1                     | 0.20               |
| 2                     | 0.45               |
| 3                     | 0.32               |

Check:  $0.03 + 0.20 + 0.45 + 0.32 = 1.00$

### 3. Mean (Expected Value)

Formula:

$$\begin{aligned}
 E(x) &= \sum x P(x) \\
 &= (0 \times 0.03) + (1 \times 0.20) + (2 \times 0.45) + (3 \times 0.32) \\
 &= 0 + 0.20 + 0.90 + 0.96 = 2.06
 \end{aligned}$$

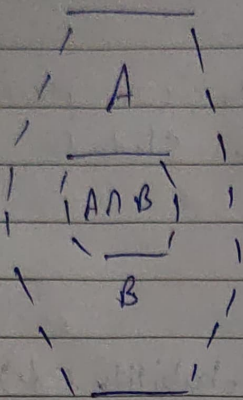
### 4. Variance

Formula:

$$\begin{aligned}
 \text{Var}(x) &= E(x^2) - [E(x)]^2 \\
 E(x^2) &= 0 + (1^2 \times 0.20) + (2^2 \times 0.45) + (3^2 \times 0.32) \\
 &= 0 + 0.20 + 1.80 + 2.88 = 4.88 \\
 \text{Var}(x) &= 4.88 - (2.06)^2 \\
 &= 4.88 - 4.24 = 0.64
 \end{aligned}$$

Ans: Variance = 0.64

Q.4) Venn Diagram in probability.



Q.5) Contingency Table and Probability Calculation

Contingency Table

| Group Discussion | Pass | Fail | Total |
|------------------|------|------|-------|
| Yes              | 90   | 20   | 110   |
| No               | 78   | 12   | 90    |
| Total            | 138  | 62   | 200   |

Q

1. Total probability

$$P(\text{Discussion and pass}) = \frac{90}{200} = 0.45$$

Ans = 0.45

2. Marginal probability

$$P(\text{Pass}) = \frac{138}{200} = 0.69$$

Ans = 0.69

## 3) Conditional probability

$$P(\text{Pass} | \text{Discussion}) = \frac{30}{110} = 0.82$$

Ans = 0.82

## Q.68 Understanding Relationships.

Interpretation.

Students who participate in group discussions have an 82% probability of passing the final exam.

Relationship

The events "Participating" affects the probability of passing. They are not mutually exclusive since a student can both participate in group discussion and pass the exam.

## Q.71 Bayes' Theorem Application.

Given,

- $P(\text{High Attendance} | \text{Pass}) = 0.70$
- $P(\text{High Attendance} | \text{Fail}) = 0.40$
- $P(\text{High Attendance}) = 0.60$

Formula,

$$P(\text{Pass} | \text{High Attendance}) = \frac{P(\text{High Attendance} | \text{Pass}) \times P(\text{Pass})}{P(\text{High Attendance})}$$

Assume:

$$P(\text{Pass}) = 0.69$$

Calculation:

$$P(\text{Pass} | \text{High Attendance}) = \frac{0.70 \times 0.69}{0.66} = \frac{0.483}{0.66} = 0.805$$

Ans:

$$P(\text{Pass} | \text{High Attendance}) = 0.805 \approx 80.5\%$$

Interpretation:

A student with high attendance (>80%) has approximately an 80.5% chance of passing the final exam.

⇒ Final Summary:

- Students who study more than 10 hrs per week are more likely to pass.
- Students with attendance above 80% have a higher probability of passing.
- Group discussion participation positively influences exam performance.
- Previous Test Scores are a strong indicator of final Exam Success.
- According to Bayes' Theorem, students with high Attendance have an 80.5% probability of passing.
- Therefore, Study hours, attendance, group discussion, and previous test score are the key factors affecting the probability of passing the Gujarat mathematics exam.